Graph Machine Learning Seminar

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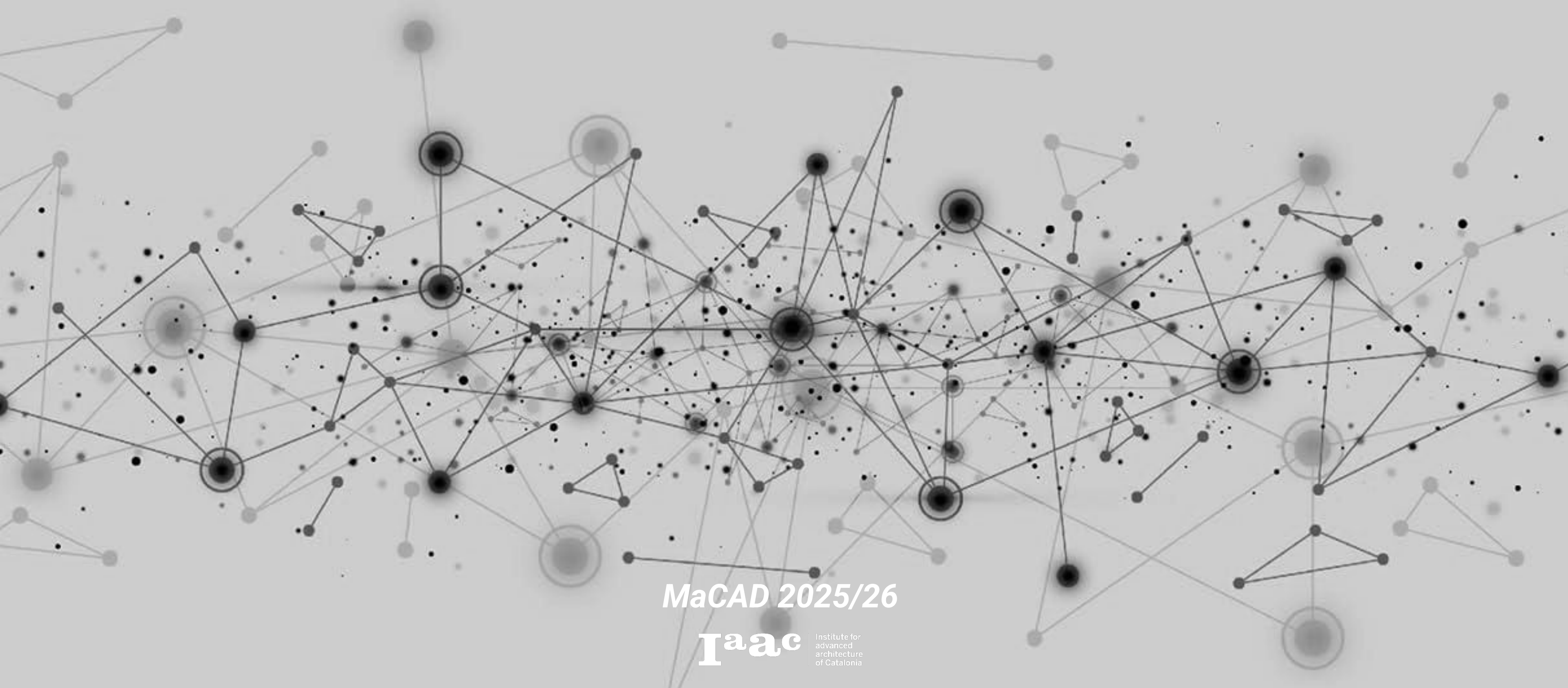
Graph Machine Learning: Learn about the latest advancements in graph data to build robust machine learning models



MaCAD 2025/26

Iaac Institute for advanced architecture of Catalonia

Assignment 01: Graph Generation from an Architectural Model

Project to analyse Iaac ACESD Studio project: Floating Grounds

Assignment 03 : Graph Machine Learning for Architectural Data
Creating BGR Graph

What This Assignment Is About?

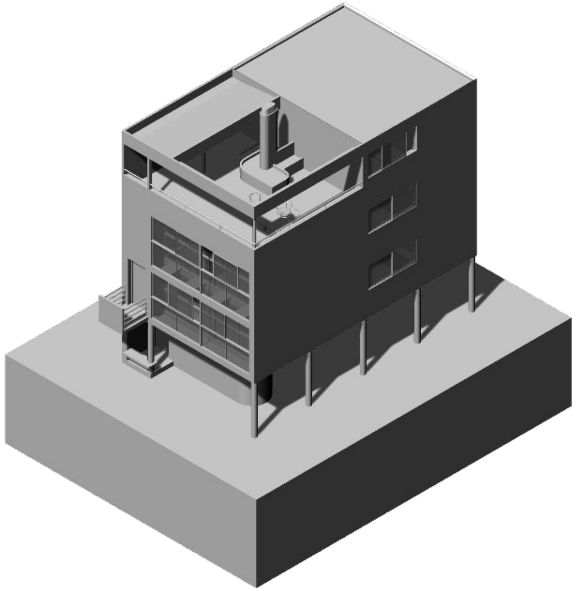
How to understand Machine Learning through Architecture.
The Data set for this assignment is called BGR (Building-Ground Relationship) and from the Notebook it classifies how a building sits on the ground into 5 types:

0 Separation (building floats above ground)
1 Separation with Plinth
2 Adherence (building sits directly on ground)
3 Adherence with Plinth
4 Interlock (building is partially embedded in ground)

Model Choice

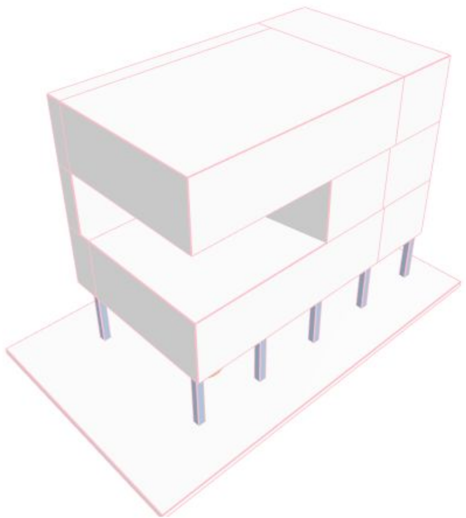
For this Assignment we find adequate to analyse one of the buildings of **Le Corbusier**, for it interesting relationship with the ground and in particularly **Maison Citrohan (1922)**.

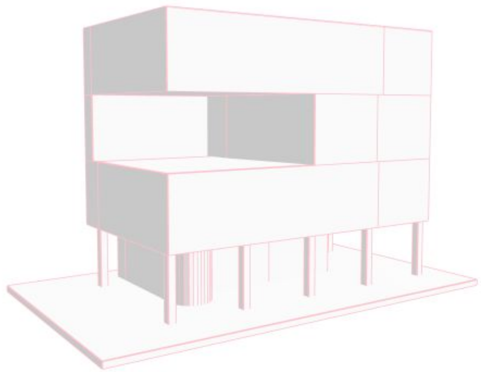
The building sits on pilotis, so it is lifted off the ground, which is separation with Plinth



Creating BGR Graph

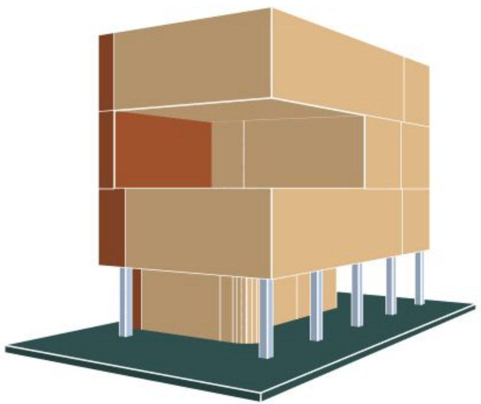
Model has a double Height that we simplified and we left the curved entrance in the ground floor just to maintain the Le Corbusier Style

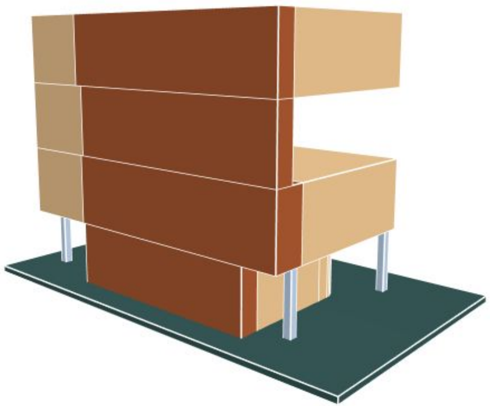




Creating BGR Graph

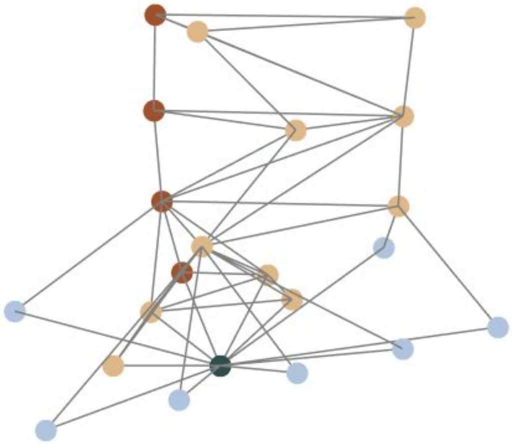
We assigned Colors to each Cell





Creating BGR Graph

We generated the adjacency Graph, each node is a cell, coloured by type; each edge represents a shared face between adjacent cells



Prediction

The model assigned 100% probability to Separation with Plinth, correctly identifying the building type.

Labels:

- 0: "Separation"
- 1: "Separation with Plinth"
- 2: "Adherence"
- 3: "Adherence with Plinth"
- 4: "Interlock"

6. Predict the dataset

```python
pred_results = pyg_2.Predict()
predicted_values = pred_results['pred'].tolist()
predicted_labels = [MAPPING[pv] for pv in predicted_values]
actual_values = pred_results['y_true'].tolist()
actual_labels = [MAPPING[av] for av in actual_values]
probabilities = pred_results['prob'].tolist()

df = pd.DataFrame({
    "Actual Value": actual_values,
    "Predicted Value": predicted_values,
    "Actual Label": actual_labels,
    "Predicted Label": predicted_labels,
    "Confidence": [round(max(p), 2) for p in probabilities]
})

df
```

[21] ✓ 0.0s Python

	Actual Value	Predicted Value	Actual Label	Predicted Label	Confidence
0	1	1	Separation with Plinth	Separation with Plinth	1.0